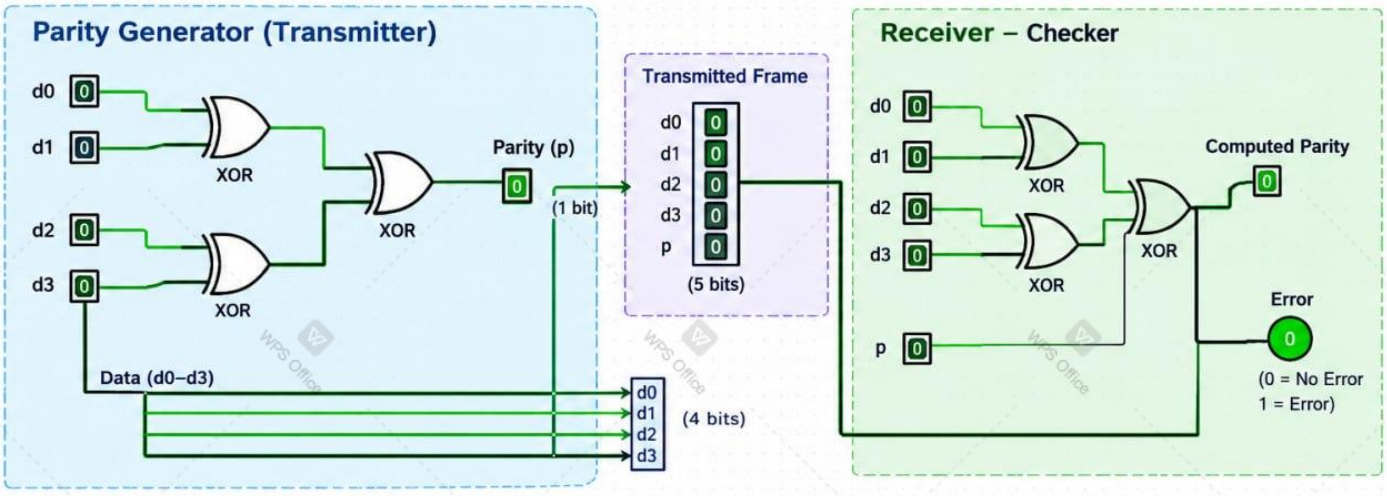
XOR Error Detection System – Logisim Diagram



Test Inputs (Logisim Switches)

d0 0
d1 0
d2 0
d3 0

Toggle these switches to test different data and error cases.

Example Frame (5 bits)

0	1	0	1	0
d0	d1	d2	d3	p

- ✓ 4-input XOR tree generates parity
- ✓ Receiver recomputes parity and compares
- ✓ Error = 1 when parity mismatch occurs
- ✓ Works for single-bit error detection